

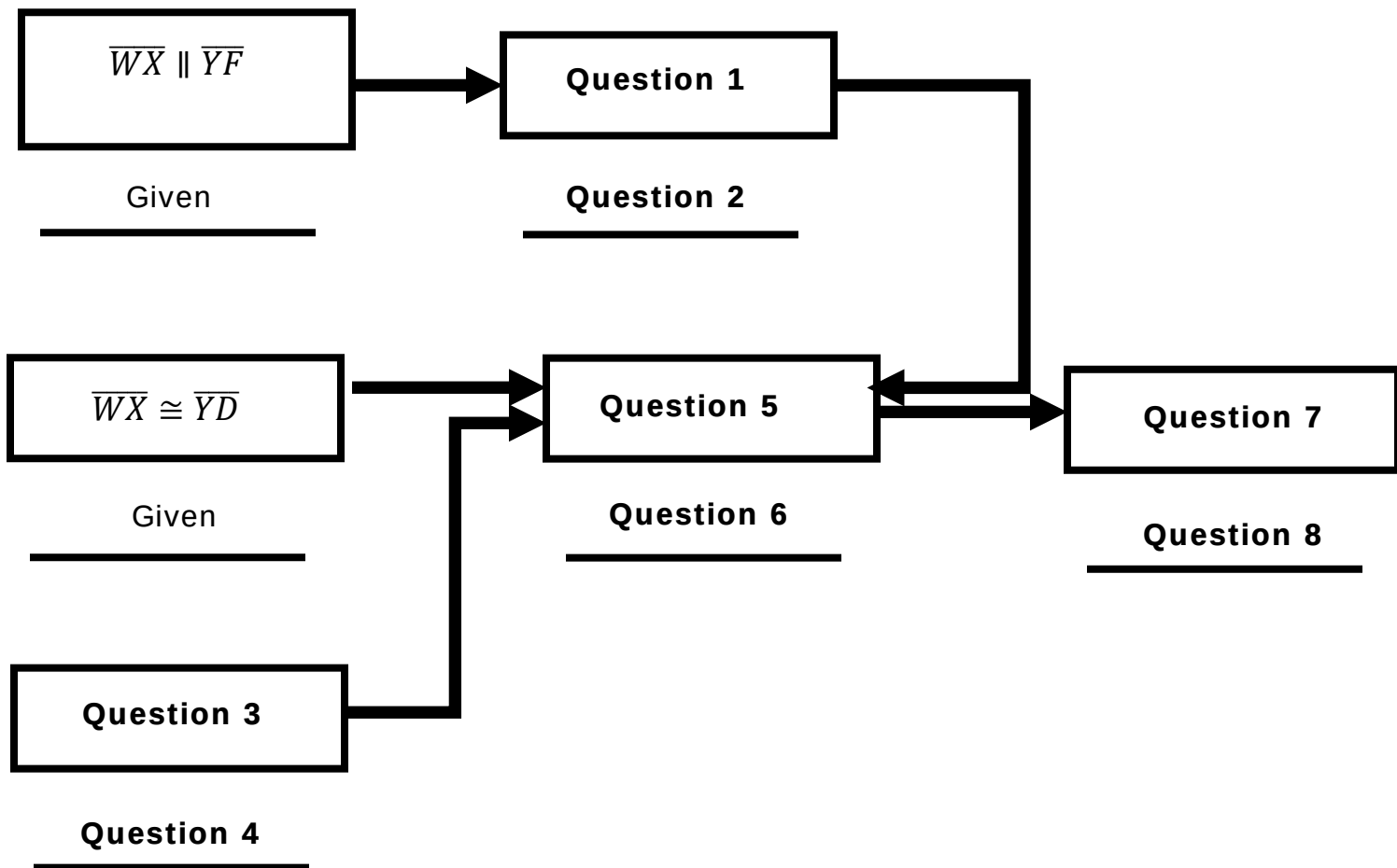
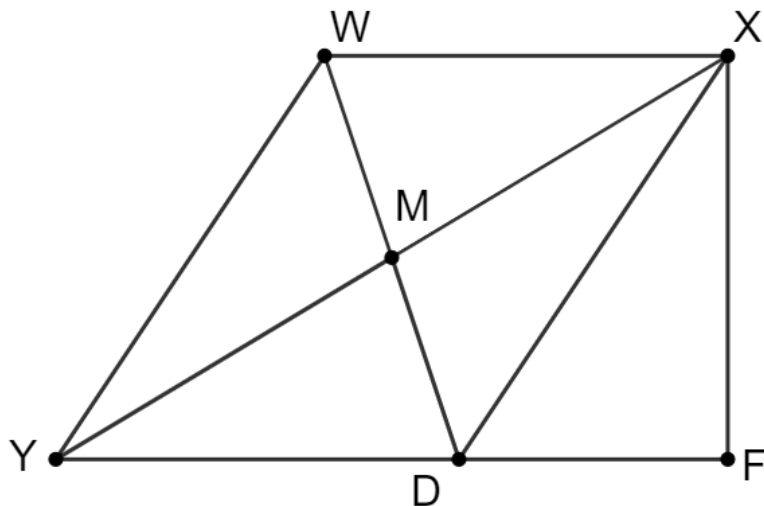
Geometry
Unit 5
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

Quadrilateral $WXYF$ is shown where
 $\overline{WX} \cong \overline{YD}$ and $\overline{WX} \parallel \overline{YF}$.

Given: $\overline{WX} \parallel \overline{YF}$, $\overline{WX} \cong \overline{YD}$
Prove: $\overline{WY} \cong \overline{XD}$

Use the given information to answer
the questions about the incomplete
proof that is shown. Each empty box
has the corresponding question
number.



1. What is the statement that belongs in the box?

$$\angle WXY \cong \angle DYX$$

2. What is the reason for the statement?

If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

3. What is the statement that belongs in the box?

$$\overline{XY} \cong \overline{XY}$$

4. What is the reason for the statement?

Reflexive Property of Congruence

5. What is the statement that belongs in the box?

$$\triangle WXY \cong \triangle DYX$$

6. What is the reason for the statement?

SAS Congruence

7. What is the statement that belongs in the box?

$$\overline{WY} \cong \overline{XD}$$

8. What is the reason for the statement?

Corresponding Parts of Congruent Triangles are Congruent

9. Explain if it is possible to prove that $\overline{WM} \cong \overline{MD}$ using the same given statements.

Yes. Since $\overline{WX} \parallel \overline{YF}$ it can be said that $\angle WXM \cong \angle DYM$ and $\angle XWM \cong \angle YDM$ because if two parallel lines are intersected by a transversal, the alternate interior angles are congruent. Therefore $\triangle WMX \cong \triangle DYM$ by ASA $\cong$. It follows that $\overline{WM} \cong \overline{MD}$ because Corresponding Parts of Congruent Triangles are Congruent.

10. Explain if it is possible to prove $\triangle WMX \cong \triangle DMY$ using the same given statements.
Yes. Since $\overline{WX} \parallel \overline{YF}$ it can be said that $\angle WXM \cong \angle DYM$ and $\angle XWM \cong \angle YDM$ because if two parallel lines are intersected by a transversal, the alternate interior angles are congruent. Therefore $\triangle WMX \cong \triangle DYM$ by ASA $\cong$.